EX:2

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SUBJECT: PROMPT ENGINEERING

SLOT: T1-S5

PASSAGE

BASICS OF BLOCKCHAIN TECHNOLOGY

Blockchain is a revolutionary technology that functions as a shared, immutable digital ledger. The name "blockchain" comes from its structure data is organized in blocks, with each new block linked to the one before it, forming a continuous chain.

Each block contains crucial data, such as a list of transactions, a timestamp, and a unique identifier called a cryptographic hash. This hash is generated from the block's contents and the hash of the previous block, ensuring that each block is tightly connected to the one before it.

How does Blockchain Technology Work?

One of the famous use of Blockchain is Bitcoin. Bitcoin is a cryptocurrency and is used to exchange digital assets online. Bitcoin uses cryptographic proof instead of third-party trust for two parties to execute transactions over the Internet. Each transaction protects through a digital signature.

Blockchain Decentralization

There is no Central Server or System which keeps the data of the Blockchain. The data is distributed over Millions of Computers around the world which are connected to the Blockchain. This system allows the Notarization of Data as it is present on every Node and is publicly verifiable.

Blockchain nodes

A node is a computer connected to the Blockchain Network. Node gets connected with Blockchain using the client. The client helps in validating and propagating transactions onto the Blockchain. When a computer connects to the Blockchain, a copy of the Blockchain data gets downloaded into the system and the node comes in sync with the latest block of data on Blockchain. The Node connected to the Blockchain which helps in the

execution of a Transaction in return for an incentive is called Miners.

Disadvantages of the current transaction system:

- Cash can only be used in low-amount transactions locally.
- The huge waiting time in the processing of transactions.
- The need for a third party for verification and execution of Transactions makes the process complex.
- If the Central Server like Banks is compromised, the whole system is affected including the participants.

Building trust with Blockchain: Blockchain network. Blockchain builds trust through the following five attributes:

- **Distributed:** The distributed ledger is shared and updated with every incoming transaction among the nodes connected to the Blockchain. All this is done in real time as there is no central server controlling the data.
- **Secure:** There is no unauthorized access to Blockchain made possible through Permissions and Cryptography.
- **Transparent:** Because every node or participant in Blockchain has a copy of the Blockchain data, they have access to all transaction data. They themselves can verify the identities without the need for mediators.
- **Consensus-based:** All relevant network participants must agree that a transaction is valid. This is achieved through the use of consensus algorithms.
- **Flexible:** Smart Contracts which are executed based on certain conditions can be written into the platform. Blockchain Networks can evolve in pace with business processes.

Application of Blockchain

- Leading Investment Banking Companies like Credit Suisse, JP Morgan Chase, Goldman Sachs, and Citigroup have invested in Blockchain and are experimenting to improve the banking experience and secure it.
- Barclays uses Blockchain to streamline the Know Your Customer (KYC) and Fund Transfer processes while filling patents against these features.
- Visa uses Blockchain to deal with business-to-business payment services.
- DHL and Accenture work together to track the origin of medicine until it reaches the consumer.
- Pfizer, an industry leader, has developed a blockchain system to keep track of and manage the inventory of medicines.
- The government of Dubai looking forward to making Dubai the first-ever city to rely on entirely and work using blockchain, even in their government office.

Future Scope of Blockchain Technology

The following are some potential uses for blockchain in the future:

- Digital Identity: Blockchain-based digital IDs might be used to store personal data safely and securely as well as offer a means of establishing identity without the need for a central authority.
- Smart Contracts: A variety of legal and financial transactions could be automated using smart contracts, self-executing contracts with the terms of the agreement put straight into lines of code.
- Decentralized Finance (DeFi): Using blockchain technology, decentralized financial systems might be built that support peer-to-peer transactions and do away with conventional intermediaries like banks.

In general, blockchain technology is still in its early stages and has a wide range of potential applications.

TYPES OF PROMPTS AND ITS RESPONSES IN DIFFERENT AI MODELS

- ZERO SHOT PROMPT:

"Summarize the following passage on blockchain technology in 150-200 words, focusing on its core concepts, how it works, advantages over traditional systems, key attributes, applications, and future scope. Keep it concise, structured, and objective.

[PASSAGE]"

1. CHATGPT:

Summary of Blockchain Technology (150–200 words)

Blockchain is a decentralized and immutable digital ledger that records transactions in a secure and transparent manner. Data is stored in **blocks**, each containing transaction details, a timestamp, and a **cryptographic hash** that links it to the previous block, forming a continuous chain. This structure ensures data integrity because altering one block would affect all subsequent blocks.

Blockchain works through a distributed network of computers called **nodes**, where each node maintains a copy of the ledger. Transactions are verified using **cryptographic techniques** and **digital signatures**, eliminating the need for third-party intermediaries. For example, Bitcoin uses blockchain to enable secure peer-to-peer digital transactions. Some nodes, called **miners**, validate transactions and add new blocks to the chain.

Compared to traditional systems that rely on centralized authorities like banks, blockchain reduces processing delays, enhances security, and minimizes risks associated with central server failures. Its key attributes include **decentralization, security, transparency, consensus-based validation, and flexibility through smart contracts**.

Blockchain has applications in banking, supply chain management, healthcare, and government services. Companies such as Visa and DHL are exploring its use. In the future, blockchain may support **digital identity systems, smart contracts, and**

decentralized finance (DeFi), offering transformative potential across multiple industries.

2. CLAUDE:

Blockchain Technology: A Summary

Blockchain is a decentralized, immutable digital ledger that stores data in linked blocks, each containing transaction records, timestamps, and cryptographic hashes connecting it to the previous block. This structure ensures data integrity across the entire chain.

The technology operates without a central server — data is distributed across millions of nodes worldwide. Each node maintains a full copy of the ledger, enabling public verification. Miners validate transactions and are rewarded with incentives for doing so.

Compared to traditional systems, blockchain eliminates reliance on third-party intermediaries, reduces processing delays, and removes single points of failure that plague centralized institutions like banks.

Trust is built through five core attributes: distributed ledger updates, cryptographic security, full transparency, consensus-based validation, and flexibility through smart contracts.

Real-world applications span across industries — major banks (JP Morgan, Goldman Sachs), payment networks (Visa), logistics (DHL), pharmaceuticals (Pfizer), and even government initiatives like Dubai's city-wide blockchain adoption.

Looking ahead, blockchain holds significant promise in digital identity management, automated smart contracts, and decentralized finance (DeFi) — though the technology remains in its early stages of mainstream adoption.

3. PERPLEXITY:

Core Concepts

Blockchain is a shared, immutable digital ledger where data is stored in sequentially linked blocks. Each block includes transactions, a timestamp, and a cryptographic hash derived from its contents and the previous block's hash, ensuring tamper-proof connectivity.

How It Works

Popularized by Bitcoin—a cryptocurrency for secure online asset exchanges—blockchain enables peer-to-peer transactions via digital signatures and cryptographic proof, bypassing third-party trust. It's decentralized across millions of global

computers (nodes). Nodes sync by downloading the full ledger via clients, validate/propagate transactions, and miners execute them for incentives.

Advantages Over Traditional Systems

Unlike cash (limited to local low-value use), banks (slow processing, third-party dependency, single-point failure risk), blockchain offers fast, verifiable notarization without central servers.

Key Attributes Building Trust

Distributed: Real-time shared ledger across nodes.

Secure: Permissions and cryptography prevent unauthorized access.

Transparent: All nodes access full transaction history for self-verification.

Consensus-based: Transactions validated via network agreement algorithms.

Flexible: Supports smart contracts for conditional automation.

Applications

Banks like JP Morgan and Barclays use it for secure transfers and KYC. Visa handles B2B payments; DHL/Accenture and Pfizer track pharmaceuticals; Dubai aims for full government blockchain integration.

Future Scope

Potential in digital IDs (secure, authority-free), smart contracts (automated legal/financial deals), and DeFi (peer-to-peer finance without banks). Still emerging with vast applications.

- FEW SHOT PROMPT:

"Summarize the following passage in 150-200 words, using a structured format: 1) Key concepts, 2) How it works, 3) Advantages, 4) Applications & future.

Example 1:

Passage: [Short example passage on AI ethics]

Summary: 1) Key concepts: Bias in models... 2) How it works: Training data filters...

Example 2:

Passage: [Short example passage on quantum computing]

Summary: 1) Key concepts: Qubits enable superposition... 2) How it works: Entanglement...

Now summarize this passage:

[PASSAGE]"

1. CHATGPT:

1) Key Concepts:
Blockchain is a **distributed digital ledger** that records transactions securely and permanently. Data is stored in **blocks**, each containing transaction details, a timestamp, and a **cryptographic hash** linking it to the previous block. This structure forms a chain that is difficult to alter. The system operates **decentrally**, meaning there is no central authority or server controlling the data. Instead, the ledger is shared across many computers called **nodes**, which store copies of the blockchain. Trust is built through five major attributes: **distribution, security, transparency, consensus, and flexibility** (including smart contracts).

2) How It Works:
When a transaction occurs (e.g., in Bitcoin), it is verified using **cryptographic digital signatures**. The transaction is broadcast to network nodes, which validate it using consensus algorithms. Once verified, it is grouped into a block and added to the blockchain. Nodes synchronize with the latest version of the ledger, while **miners** help validate transactions and receive incentives.

3) Advantages:
Blockchain reduces dependence on intermediaries, increases transparency, improves security through cryptography, enables faster verification, and prevents single-point failures seen in centralized systems.

4) Applications & Future:
Blockchain is used in **banking, payments, supply chains, healthcare, and**

government systems. Future potential includes **digital identity systems, smart contracts, and decentralized finance (DeFi)**, which could automate transactions and remove traditional financial intermediaries.

2. CLAUDE:

1) Key Concepts Blockchain is a distributed, immutable digital ledger where data is stored in cryptographically linked blocks. Each block contains transaction records, a timestamp, and a unique hash tied to the previous block. Decentralization means no central server controls the data — it is distributed across millions of nodes worldwide.

2) How It Works When a transaction occurs, it is verified through consensus algorithms requiring agreement from network participants. Nodes download and sync the full blockchain, while miners validate transactions in exchange for incentives. Cryptographic digital signatures secure each transaction, eliminating the need for third-party intermediaries like banks.

3) Advantages Blockchain builds trust through five core attributes: distributed storage, cryptographic security, transparency, consensus-based validation, and flexibility via smart contracts. This addresses traditional system weaknesses such as slow processing, central server vulnerability, and reliance on intermediaries.

4) Applications & Future Major institutions — JP Morgan, Visa, Barclays, Pfizer, and DHL — already leverage blockchain for payments, KYC, and supply chain tracking. Dubai aims to become the first fully blockchain-powered city. Future potential includes decentralized finance (DeFi), smart contract automation of legal agreements, and secure blockchain-based digital identity systems.

3. PERPLEXITY:

1. **Key concepts:** Blockchain is a shared, immutable digital ledger structured as chained blocks, each containing transactions, timestamps, and cryptographic hashes linking to the previous block. It operates in a decentralized network of nodes (computers) without a central server, using cryptography for security, digital signatures for transactions (e.g., Bitcoin), and consensus algorithms for validation. Core attributes include distribution, security, transparency, consensus, and flexibility via smart contracts.
2. **How it works:** Transactions are signed digitally, validated by nodes (including miners who earn incentives), and propagated across the network. New blocks are added only after consensus, ensuring real-time updates and public verifiability. Nodes sync by downloading the full blockchain copy, making data tamper-proof due to hash linkages.
3. **Advantages:** Eliminates third-party intermediaries, reduces transaction delays and complexity, enhances security against hacks (no single point of failure), provides transparency for self-verification, and builds trust through decentralization—addressing issues like cash limitations and central server vulnerabilities.

4. **Applications & future:** Current uses include banking (e.g., JP Morgan, Barclays for KYC/transfers), payments (Visa), supply chain tracking (DHL, Pfizer), and government initiatives (Dubai). Future scope: digital identities, automated smart contracts, DeFi for peer-to-peer finance, and broader automation across industries as the technology matures.

- ROLE BASED PROMPT:

"You are an expert technical writer for IEEE Spectrum magazine. Write a 150-200 word summary of the following passage on blockchain technology. Structure it with clear sections on basics, mechanics, trust attributes, real-world applications, and future potential. Use precise language suitable for engineering professionals. [PASSAGE]"

1. CHATGPT:

Basics

Blockchain is a distributed digital ledger designed to record transactions in a tamper-resistant manner. Data are grouped into sequential blocks containing transaction records, timestamps, and cryptographic hashes. Each block references the hash of the previous block, creating a chained structure that prevents retroactive modification without altering the entire ledger.

Mechanics

Transactions are broadcast to a decentralized network of nodes. Each node maintains a synchronized copy of the ledger and participates in validating transactions through cryptographic verification and consensus protocols. Systems such as **Bitcoin** demonstrate this model by using digital signatures and cryptographic proof rather than centralized intermediaries. Specialized nodes, often called miners, verify transactions and append validated blocks to the chain.

Trust Attributes

Blockchain establishes trust through several properties: distributed data replication, strong cryptographic security, transparency across nodes, consensus-based validation, and programmable logic via smart contracts. These characteristics reduce dependence on centralized authorities and improve auditability.

Real-World Applications

Financial institutions including **JPMorgan Chase**, **Goldman Sachs**, and **Citigroup** are experimenting with blockchain for payments and settlement systems. Logistics and healthcare firms such as **DHL** and **Pfizer** employ blockchain to track pharmaceutical supply chains and verify product provenance.

Future Potential

Emerging applications include decentralized finance platforms, blockchain-based digital identity systems, and automated contractual execution through smart contracts—indicating significant potential for secure, trustless digital infrastructure across industries.

2. ***CLAUDE***:

Blockchain Technology: A Technical Overview

Fundamentals

Blockchain is a distributed, immutable digital ledger that organizes data into cryptographically linked blocks. Each block contains transaction records, a timestamp, and a hash derived from both its own contents and the preceding block's hash, creating a tamper-evident chain.

Mechanics

The network operates across millions of nodes — computers that download and synchronize ledger copies upon joining. Nodes called miners validate and propagate transactions, receiving incentives for their computational work. No central server governs the system.

Trust Architecture

Blockchain establishes trust through five core attributes: distributed real-time updates, cryptographic access controls, participant-level transparency, consensus-based transaction validation, and programmable flexibility via smart contracts.

Current Applications

Major financial institutions — including JP Morgan Chase, Goldman Sachs, and Visa — are actively deploying blockchain for payments and compliance. In supply chain, DHL and Pfizer use it for pharmaceutical tracking. Dubai is piloting city-wide governmental blockchain integration.

Future Trajectory

Emerging applications include self-sovereign digital identity systems, automated smart contract execution for legal and financial instruments, and decentralized finance (DeFi) frameworks that eliminate traditional banking intermediaries — signaling blockchain's potential to fundamentally reshape institutional infrastructure.

3. PERPLEXITY:

Here's a 178-word summary of the passage, crafted for IEEE Spectrum readers with precise, engineering-focused language.

Basics
Blockchain is a shared, immutable digital ledger where data resides in chronologically linked blocks. Each block holds transactions, a timestamp, and a cryptographic hash derived from its contents and the prior block's hash, forming an unbreakable chain.

Mechanics
Decentralized across global nodes—computers synced via clients that validate and propagate transactions—blockchain eliminates central servers. Miners execute transactions for incentives, using digital signatures and consensus algorithms. This addresses legacy system flaws like slow processing, third-party dependency, and single-point failures.

Trust Attributes
Trust emerges from five pillars: distributed real-time updates; secure access via permissions and cryptography; transparent verification on every node; consensus-driven validation; and flexible smart contracts that automate conditional execution.

Real-World Applications
Banks like JP Morgan and Credit Suisse enhance security and KYC; Visa streamlines B2B payments; DHL/Accenture and Pfizer track pharmaceuticals; Dubai aims for full governmental blockchain integration.

Future Potential
Emerging uses include secure digital identities, automated smart contracts, and DeFi for intermediary-free peer-to-peer finance, signaling blockchain's expansive evolution.

- CHAIN-OF-THOUGHT PROMPT:

"Summarize the passage on blockchain in 150-200 words. Think step-by-step:

1. Identify the main sections: basics/structure, how it works (e.g., Bitcoin, nodes), disadvantages it solves, trust attributes, applications, future scope.

2. Extract 2-3 key points per section.

3. Note connections (e.g., decentralization links to trust and apps).

4. Outline a concise structure: intro, core mechanics, benefits/apps, future.

5. Write the summary based on this outline.

[PASSAGE]

Step-by-step reasoning:

[Model will fill this]

Final summary:"

1. CHATGPT:

Final Summary (150 – 200 words)

Blockchain is a distributed digital ledger technology designed to record transactions securely and immutably. Data are stored in sequential blocks containing transaction records, timestamps, and cryptographic hashes. Each block references the previous one, forming a chain that prevents unauthorized modification and ensures data integrity.

Blockchain operates through a decentralized network of nodes—computers that maintain synchronized copies of the ledger and validate transactions. A prominent implementation is **Bitcoin**, which enables peer-to-peer digital asset exchange using cryptographic proofs and digital signatures instead of centralized intermediaries. This architecture addresses limitations of traditional transaction systems, such as reliance on third parties, long processing times, and vulnerability to centralized server failures.

Trust in blockchain networks arises from key attributes: distributed data storage, strong cryptographic security, transparency across participants, consensus-based validation mechanisms, and programmable smart contracts that automate agreements.

Industries are actively experimenting with blockchain applications. Financial institutions like **JPMorgan Chase** and **Goldman Sachs** explore secure financial transactions, while organizations such as **DHL** and **Pfizer** use it for supply-chain tracking and pharmaceutical management.

Looking ahead, blockchain may enable secure digital identity systems, automated smart contracts, and decentralized financial ecosystems, indicating substantial long-term potential for secure, transparent digital infrastructure.

2. CLAUDE:

Step-by-step reasoning:

- **Main sections identified:** Structure/basics, decentralized mechanics, trust attributes, problems solved, real-world applications, future potential.
- **Key points per section:**

Basics: Immutable linked blocks, cryptographic hashes, tamper-evident chain

Mechanics: Distributed nodes, miners validate transactions, no central server

Trust: Distributed, secure, transparent, consensus-based, flexible

Problems solved: Eliminates third-party dependency, reduces delays, prevents single-point failure

Applications: Finance (JP Morgan, Visa), pharma supply chain (DHL, Pfizer), governance (Dubai)

Future: Digital identity, smart contracts, DeFi

- **Connections:** Decentralization directly enables trust attributes, which in turn makes diverse real-world applications viable and DeFi's future feasible.
- **Outline:** Intro → Core mechanics → Trust/benefits → Current applications → Future scope

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Final Summary:

Blockchain is a distributed, immutable digital ledger where data is stored in cryptographically linked blocks, each containing transactions, timestamps, and a unique hash tying it to the previous block. Operating across millions of nodes with no central authority, the system uses miners to validate transactions, eliminating dependence on intermediaries like banks while reducing delays and single-point vulnerability.

Trust is established through five attributes: distributed real-time updates, cryptographic security, participant-level transparency, consensus-based validation, and programmable smart contracts. These properties make blockchain highly resilient and adaptable.

Applications are already widespread — JP Morgan, Visa, and Barclays deploy it for payments and compliance; DHL and Pfizer use it for pharmaceutical tracking; Dubai is piloting full governmental integration.

Looking ahead, blockchain's greatest promise lies in decentralized digital identity, automated legal and financial smart contracts, and peer-to-peer DeFi systems that could fundamentally replace traditional banking infrastructure — though the technology remains in its early, rapidly evolving stages.

3. PERPLEXITY:

Step-by-Step Reasoning

Main sections identified: Basics/structure (ledger, blocks, hashes); How it works (Bitcoin, decentralization, nodes/miners); Disadvantages solved (cash limits, delays, third-parties, central failures); Trust attributes (distributed, secure, transparent, consensus-based, flexible); Applications (banking, payments, tracking, government); Future scope (digital IDs, smart contracts, DeFi).

Key points per section:

Basics: Shared immutable ledger; blocks with transactions/timestamps/hashes linked sequentially.

How it works: Bitcoin for crypto exchanges via signatures; no central server; nodes sync/validate; miners incentivized.

Disadvantages solved: Local cash limits; slow/complex third-party processes; single-point failures.

Trust attributes: Distributed real-time updates; crypto security; full transparency; consensus validation; smart contract flexibility.

Applications: Banks (JP Morgan/Barclays KYC); Visa payments; DHL/Pfizer tracking; Dubai government.

Future: Secure digital IDs; automated contracts; intermediary-free DeFi.

Connections noted: Decentralization underpins trust attributes (e.g., distributed ledger enables transparency/security), solves traditional flaws, and enables apps/future uses like DeFi/smart contracts.

Outline: Intro (core concepts); Core mechanics (working/decentralization); Benefits (attributes addressing disadvantages); Apps/future.

Summary written below based on outline.

Final Summary

Introduction to Core Concepts
Blockchain is a revolutionary, shared immutable ledger of linked blocks, each holding transactions, timestamps, and hashes tying to the prior block for integrity.

Core Mechanics
Pioneered in Bitcoin for trustless cryptocurrency exchanges via digital signatures, it operates decentralized across global nodes. Nodes sync the full chain, validate transactions; miners process them for rewards, enabling notarization without central servers.

Benefits and Trust Attributes
It overcomes traditional pitfalls like cash's local limits, banking delays/third-party complexity, and server vulnerabilities. Trust stems from: distributed real-time ledgers, cryptographic security, node-level transparency, consensus algorithms, and flexible smart contracts.

Applications and Future Scope
Used by JP Morgan/Barclays (KYC/transfers), Visa (B2B), DHL/Pfizer (pharma tracking), and Dubai (government ops). Future holds digital IDs, self-executing contracts, and DeFi for peer-to-peer finance—still early-stage with vast potential.

(Word count: 168)

COMPARISON TABLE

AI MODEL	ACCURACY	COHERENCE	SIMPLICITY	SPEED	USER EXPERIENCE
CHATGPT	80%	85%	80%	85%	95%
CLAUDE	90%	83%	85%	80%	100%
PERPLEXITY	80%	80%	90%	95%	90%